

# Review of Multiple Linear Regression

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# Road Map for Today

- 1 Review of multiple linear regression
  - coefficient interpretation;
  - qualitative predictors;
  - interaction terms.
- 2 Quick R review
- 3 Example using the Credit Card data
  - fitting and interpreting regression models;
  - generating predictions;
  - coding exercise.

# Motivating Example

Today we will be using the credit card data referenced in the ISL textbook.

This dataset consists of information on 400 customers.

There are 11 variables, 7 of which are **quantitative** and 5 of which are **qualitative**.

**Qualitative predictors** take on a discrete set of values.

Today, we will be working with the variables: 'Married', 'Region', 'Student', 'Income'

For more information about the dataset:

<https://rdrr.io/cran/ISLR2/man/Credit.html>

# Qualitative Predictors with Two Levels

Suppose we want to study the difference in credit-card Balance between married and single individuals, ignoring the other variables.

We create a dummy variable:

$$X_i = \begin{cases} 1, & \text{if the } i\text{th person is married,} \\ 0, & \text{if the } i\text{th person is single.} \end{cases}$$

The resulting model:

$$Y_i = \beta_0 + \beta_1 X_i + \epsilon_i = \begin{cases} \beta_0 + \beta_1 + \epsilon_i, & \text{if } i\text{th person is married,} \\ \beta_0 + \epsilon_i, & \text{if } i\text{th person is single.} \end{cases}$$

# Qualitative Predictors with More Than Two Levels

With more than two levels, we create additional dummy variables. For example, for the Region variable we create two dummy variables. They could be

$$X_{i1} = \begin{cases} 1, & \text{if the } i\text{th person lives South,} \\ 0, & \text{otherwise,} \end{cases}$$

and

$$X_{i2} = \begin{cases} 1, & \text{if the } i\text{th person lives West,} \\ 0, & \text{otherwise.} \end{cases}$$

# Qualitative Predictors with More Than Two Levels

The two dummy variables can be included in the regression model:

$$Y_i = \beta_0 + \beta_1 X_{i1} + \beta_2 X_{i2} + \epsilon_i = \begin{cases} \beta_0 + \beta_1 + \epsilon_i, & \text{if } i\text{th person lives South,} \\ \beta_0 + \beta_2 + \epsilon_i, & \text{if } i\text{th person lives West} \\ \beta_0 + \epsilon_i, & \text{if } i\text{th person lives East..} \end{cases} .$$

There is always one fewer dummy variable than the number of levels.

The level for which all dummy variables equal zero is called the **baseline**.

# Motivating Example

Coefficients:

|             | Estimate | Std. Error | t value | Pr(> t )   |
|-------------|----------|------------|---------|------------|
| (Intercept) | 531.00   | 46.32      | 11.464  | <2e-16 *** |
| RegionSouth | -12.50   | 56.68      | -0.221  | 0.826      |
| RegionWest  | -18.69   | 65.02      | -0.287  | 0.774      |

**Interpretation:** The West category tends to have 18.69 less debt than the East category, and the South category tends to have 12.50 less debt than the East category.

**Exercise:** What does the p-value tell us about the coefficient?

# Adding Interaction Terms

Consider the model

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \epsilon.$$

Regardless of the value of  $X_2$ , a one-unit increase in  $X_1$  is associated with a  $\beta_1$ -unit increase in  $Y$ .

Now consider a model with an interaction:

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_1 X_2 + \epsilon.$$

Equivalently,

$$Y = \beta_0 + (\beta_1 + \beta_3 X_2) X_1 + \beta_2 X_2 + \epsilon.$$

Because the coefficient of  $X_1$  depends on  $X_2$ , the effect of  $X_1$  on  $Y$  is no longer constant.

# Adding Interaction Terms

Let

$$X_{i1} = \begin{cases} 1, & \text{if the } i\text{th person is a student,} \\ 0, & \text{otherwise,} \end{cases}$$

and let  $X_{i2}$  denote Income.

Consider

$$Y_i = \beta_0 + \beta_1 X_{i1} + \beta_2 X_{i2} + \beta_3 X_{i1} X_{i2} + \epsilon_i.$$

For non-students:

$$E(Y_i \mid X_{i1} = 0) = \beta_0 + \beta_2 X_{i2}.$$

For students:

$$E(Y_i \mid X_{i1} = 1) = \beta_0 + \beta_1 + \beta_2 X_{i2} + \beta_3 X_{i2}.$$

# Adding Interaction Terms

For the model

$$Y_i = \beta_0 + \beta_1 X_{i1} + \beta_2 X_{i2} + \beta_3 X_{i1} X_{i2} + \epsilon_i,$$

where  $X_{i1}$  indicates Student status and  $X_{i2}$  is Income:

- Interpretation of  $\beta_2$ ?
- Interpretation of  $\beta_3$ ?
- Interpretation of  $\beta_2 + \beta_3$ ?

# Next Steps

R Review + Linear Regression Example!